Functional Annotation Report: rbsA-I (PP_2455 / Q88K37) in *Pseudomonas putida* KT2440

Summary

rbsA-I (PP_2455; UniProt Q88K37) encodes the cytoplasmic ATP-binding (ATPase) subunit of the high-affinity D-ribose ABC importer of *Pseudomonas putida* KT2440. It is a 524-amino-acid soluble protein built from two fused nucleotide-binding domains (NBDs), each carrying the canonical Walker A / P-loop and Walker B catalytic motifs of the ABC-transporter ATPase superfamily. Its primary function is not to catalyze a stand-alone metabolic reaction but to serve as the energy-coupling motor of a membrane transport machine: it binds and hydrolyzes ATP at the cytoplasmic face of the inner membrane, and the resulting conformational cycling drives the vectorial translocation of D-ribose from the periplasm into the cytoplasm through its partner permease. The enzyme classification EC 7.5.2.7 (previously EC 3.6.3.17), KEGG orthology K10441 ("ribose transport system ATP-binding protein"), and GO terms GO:0005524 (ATP binding), GO:0016887 (ATP hydrolysis activity), and GO:0015749 (monosaccharide transmembrane transport) all converge on this assignment.

The identity of the protein is unambiguous. PP_2455 sits within a complete, canonically organized chromosomal ribose (*rbs*) operon (PP_2454–PP_2460) that encodes the periplasmic ribose-binding protein (RbsB), the ATPase (RbsA-I), the permease (RbsC), a LacI-family repressor (RbsR), and the downstream catabolic enzymes ribokinase (RbsK), ribose pyranase (RbsD), and a ribonucleoside hydrolase. This gene neighborhood mirrors the archetypal *Escherichia coli rbsDACBK–rbsR* system, in which the RbsABC complex is the high-affinity transporter and RbsD/RbsK convert imported D-ribose into D-ribose-5-phosphate for entry into the pentose-phosphate pathway. RbsA-I shares 41% amino-acid identity with the biochemically characterized *E. coli* RbsA, a level well above the threshold generally accepted for confident cross-genus functional transfer among transporter ATPases.

The substrate specificity of the transporter — D-ribose — is determined by the periplasmic binding protein RbsB rather than by RbsA-I itself; the ATPase is the generic energizing component. RbsA-I therefore acts at the cytoplasmic face of the inner membrane, physically

docked onto the RbsC permease homodimer, where its nucleotide-binding state governs the assembly and conformational state of the full $RbsABC_2$ translocation complex. This report details the evidence for each of these conclusions, presents a mechanistic model, evaluates the supporting literature, and identifies the residual knowledge gaps and experiments that would close them.

Key Findings

Finding 1 — PP_2455 (rbsA-I) encodes the ATP-binding (ATPase) subunit of a canonical D-ribose ABC importer

The core functional assignment rests on a convergence of sequence, domain, and orthology evidence. UniProt Q88K37 describes a 524-amino-acid protein annotated with the keywords ATP-binding, Nucleotide-binding, Translocase/Hydrolase, and Sugar transport. Domain analysis identifies two copies of the Pfam PF00005 ABC_tran domain (a tandem, repeated architecture), together with InterPro signatures IPR050107 (ABC carbohydrate-import ATPase), IPR003439 (ABC transporter-like ATP-binding domain), IPR003593 (AAA+ ATPase), and IPR027417 (P-loop NTPase). The KEGG entry ppu:PP_2455 maps to KEGG Orthology K10441, "ribose transport system ATP-binding protein," with EC 7.5.2.7 (ABC-type D-ribose transporter; the modern reclassification of the legacy EC 3.6.3.17 cited in the UniProt record). The Gene Ontology annotations — GO:0005524 (ATP binding), GO:0016887 (ATP hydrolysis activity), and GO:0015749 (monosaccharide transmembrane transport) — are fully consistent with an ABC-transporter ATPase. Direct inspection of the primary sequence confirms a Walker A / P-loop motif (...TGENGAGKSTL...) diagnostic of nucleotide binding.

The protein's length is itself informative. At 524 aa, RbsA-I is roughly twice the size of a single ~260-aa nucleotide-binding domain, consistent with two fused NBDs — exactly the architecture of the experimentally defined *E. coli* RbsA. The reconstitution study of the *E. coli* ribose transporter states plainly that "the ribose transporter in *Escherichia coli* is a tripartite complex consisting of a cytoplasmic ATP-binding cassette protein, RbsA, with fused nucleotide binding domains; a transmembrane domain homodimer, $RbsC_2$; and a periplasmic substrate binding protein, RbsB" (PMID: 25533465). This description of RbsA as a cytoplasmic ABC protein with two fused NBDs matches the observed 524-aa, two-NBD architecture of Q88K37 point for point.

Interpretation: RbsA-I is the ATP-hydrolyzing engine of an importer. It supplies the free energy — through ATP binding and hydrolysis — that powers uphill accumulation of D-ribose against a concentration gradient. It is a transporter subunit, not an independent catabolic enzyme.

Finding 2 — rbsA-I lies within a complete chromosomal ribose (*rbs*) operon dedicated to D-ribose uptake and catabolism

Gene context is one of the strongest lines of evidence for the function of a transporter subunit, because ABC-importer genes are almost universally clustered with their cognate binding protein, permease, and downstream catabolic enzymes. The KEGG genomic neighborhood of PP_2455 (chromosomal coordinates ~2,801,617–2,803,191 and flanking genes) comprises:

Locus tag	Gene	Product	Role in ribose utilization
PP_2454	rbsB	Periplasmic ribose-binding protein	Captures D-ribose in periplasm; sets substrate specificity
PP_2455	**rbsA-I**	**ABC transporter ATP-binding subunit**	**Energizes transport via ATP hydrolysis (this study)**
PP_2456	rbsC	Ribose permease (transmembrane domain)	Forms the translocation channel
PP_2457	rbsR	LacI-family transcriptional repressor	Regulates operon expression
PP_2458	rbsK	Ribokinase	Phosphorylates D-ribose → D-ribose-5-phosphate
PP_2459	rbsD	Ribose pyranase	Interconverts ribose anomers/forms for kinase
PP_2460	—	Ribonucleoside hydrolase	Liberates ribose from nucleosides

This organization recapitulates the *E. coli rbsDACBK–rbsR* system, in which RbsABC is the high-affinity transporter and RbsD/RbsK convert D-ribose into D-ribose-5-phosphate. The regulatory and metabolic logic of the operon is well established: "the genes for the transport and initial-step metabolism of D-ribose form a single *rbsDACBK* operon. RbsABC forms the ABC-type high-affinity D-ribose transporter, while RbsD and RbsK are involved in the conversion of D-ribose into D-ribose 5-phosphate" (PMID: 23651393). The presence in the *P. putida* cluster of all three transporter genes (*rbsB*, *rbsA-I*, *rbsC*) together with *rbsK*, *rbsD*, and a ribonucleoside hydrolase confirms that this locus constitutes a complete, self-contained ribose acquisition-and-catabolism module.

Interpretation: The operon context removes essentially all ambiguity about the substrate and pathway. RbsA-I energizes the import of D-ribose, which is immediately handed off to RbsK/RbsD for conversion to D-ribose-5-phosphate and entry into the pentose-phosphate pathway. The co-localized RbsR repressor places the entire module under substrate-responsive transcriptional control.

Finding 3 — RbsA localizes to the cytoplasmic face of the inner membrane and powers transport via ATP hydrolysis within the $RbsABC_2$ complex

The subcellular localization annotation in UniProt places Q88K37 at the cell inner membrane / cell membrane as a peripheral protein on the cytoplasmic side. This is precisely the expected topology for a soluble ABC-ATPase that docks onto the cytoplasmic coupling helices of its permease. The functional dynamics of this arrangement were resolved biochemically for the *E. coli* orthologue. In vitro reassembly of the ribose transporter demonstrated that RbsA associates with the $RbsC_2$ permease homodimer, and that the composition of the isolated complex depends on the nucleotide state: "we were able to purify a full complex, $RbsABC_2$, in the presence of stable, transition state mimics (ATP, Mg^{2+}, and VO_4); a RbsAC complex in the presence of ADP and Mg^{2+}; and a heretofore unobserved RbsBC complex in the absence of cofactors" (PMID: 25533465).

This nucleotide-dependent assembly is direct experimental evidence that RbsA's catalytic cycle — ATP binding, hydrolysis to ADP, and product release — drives the conformational transitions that alternately assemble and disassemble the translocation-competent complex. The transition-state mimic ($ATP{\cdot}Mg{\cdot}VO_4$) traps the full $RbsABC_2$ complex, whereas the post-hydrolysis ADP·Mg state yields only RbsAC, illustrating how the ATPase gates the recruitment of the substrate-loaded binding protein. Because RbsA-I of *P. putida* is 41% identical to this *E. coli* RbsA and conserves all catalytic motifs (Finding 4), the same mechanism is confidently inferred.

Interpretation: RbsA-I performs its function as a membrane-associated (but not membrane-integral) subunit on the inner (cytoplasmic) leaflet of the inner membrane. Its ATPase cycle mechanically couples chemical energy to the alternating-access motion of the RbsC permease, achieving vectorial import of D-ribose.

Finding 4 — RbsA-I shares 41% identity with experimentally characterized *E. coli* RbsA and possesses the conserved dual ABC-ATPase motif set

A Needleman–Wunsch global alignment of Q88K37 (524 aa) against *E. coli* RbsA (UniProt P04983, 501 aa) yields 41.2% identity over 498 aligned columns. For transporter ATPases, identity in this range across genera is a strong basis for functional transfer, especially when combined with conserved active-site motifs and identical genomic context. A motif scan of Q88K37 confirms the tandem two-NBD architecture expected of RbsA:

Motif	NBD1	NBD2
Walker A / P-loop	GENGAGKS (residue ~48)	P-loop-type region (AASGLGKT, res ~18 region)
Walker B (…hhhhDE)	LILDE (res ~174)	LLFDE (res ~430)

The two cleanly separated Walker B motifs demarcate the two fused nucleotide-binding domains, exactly as expected for a "double-NBD" importer ATPase. The presence of two complete Walker A/Walker B pairs distinguishes RbsA from single-NBD ABC-ATPases (which function as homodimers) and identifies it as the intramolecularly-fused variant characteristic of the ribose importer family.

Interpretation: The high sequence identity to a functionally validated orthologue, combined with the fully conserved catalytic apparatus (two P-loops, two Walker B motifs), makes the functional assignment of RbsA-I as an ATP-hydrolyzing ribose-importer subunit essentially certain by homology, independent of the operon-context evidence.

Finding 5 — *P. putida* encodes two ribose ABC-ATPase paralogs; rbsA-I (PP_2455) is the copy embedded in the complete catabolic ribose operon

P. putida KT2440 possesses two loci mapped to KEGG Orthology K10441: PP_2455 (rbsA-I, Q88K37, 524 aa) and PP_2759 (rbsA / "rbsA-II", Q88J90, 512 aa). The two paralogs share only 42.9% identity by global alignment, comparable to their individual identity with *E. coli* RbsA, indicating an ancient duplication or independent acquisition rather than a recent gene doubling. Critically, only PP_2455 sits within a *complete* ribose operon: it is flanked by the substrate-binding protein (rbsB, PP_2454), the permease (rbsC, PP_2456), the regulator (rbsR, PP_2457), and the downstream catabolic genes ribokinase (rbsK, PP_2458), ribose pyranase (rbsD, PP_2459), and a ribonucleoside hydrolase (PP_2460). By contrast, the PP_2759 cluster (PP_2757–PP_2761) contains two K10439-type binding proteins, the PP_2759 ATPase, and two K10440-type permeases but *lacks adjacent ribokinase and pyranase genes*.

Interpretation: The absence of co-localized catabolic enzymes at the PP_2759 cluster suggests that paralog carries out a related but distinct transport role (potentially a different pentose or ribose-derivative, or feeding a different metabolic branch), whereas rbsA-I (PP_2455) is the transporter ATPase directly wired to D-ribose catabolism through the pentose-phosphate pathway. This paralog distinction is important for correct annotation: RbsA-I is the "canonical," catabolically-coupled ribose importer ATPase of *P. putida*.

Mechanistic Model / Interpretation

The findings assemble into a coherent picture of a Type I ABC importer operating at the *P. putida* inner membrane. The transport module is tripartite plus a soluble receptor, and RbsA-I is its ATP-hydrolyzing motor.

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                  PERIPLASM
         D-ribose  ●
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            ┌────────▼────────┐
            │  RbsB (PP_2454) │   periplasmic binding protein
            │  captures ribose│   → sets D-ribose specificity
            └────────┬────────┘
                     │ delivers substrate
    ═════════════════╪═════════════════════════  INNER MEMBRANE
                     │
  ┌──────────────┐           ┌──────────────┐
  │  RbsC (PP_2456) permease homodimer (RbsC2) │  translocation channel
  └──────┬───────┘           └──────┬───────┘
         │  coupling helices         │
  ┌──────▼────────────────────────────▼──────┐
  │        RbsA-I (PP_2455)  —  ATPase        │  CYTOPLASMIC face
  │   NBD1 [Walker A GENGAKGS | Walker B LILDE]│
  │   NBD2 [Walker A/P-loop   | Walker B LLFDE]│
  │        ATP → ADP + Pi  (EC 7.5.2.7)        │
  └──────────────────┬─────────────────────────┘
                  CYTOPLASM
         D-ribose  ●
                   │  RbsK ribokinase (PP_2458)
                   │  RbsD pyranase   (PP_2459)
                   ▼
         D-ribose-5-phosphate  →  Pentose-phosphate pathway
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Transport cycle. In the resting state, the periplasmic RbsB scavenges D-ribose. Substrate-loaded RbsB docks onto the periplasmic face of the $RbsC_2$ permease. On the cytoplasmic side, RbsA-I binds two molecules of Mg·ATP at the interface of its two NBDs. ATP binding drives NBD dimerization/closure, which is transmitted through the permease coupling helices to open the transmembrane cavity toward the periplasm and accept the substrate. ATP hydrolysis and Pi/ADP release reset the transporter to the inward-open state, delivering D-ribose to the cytoplasm. The *E. coli* reconstitution study captured discrete snapshots of this cycle — full $RbsABC_2$ in the ATP transition-state mimic, RbsAC in the ADP state — providing direct structural-biochemical validation of the nucleotide-gated mechanism (PMID: 25533465).

Metabolic coupling. Imported D-ribose is immediately committed to catabolism by the co-operonic enzymes: ribose pyranase (RbsD) equilibrates the ribose anomeric forms, and ribokinase (RbsK) phosphorylates D-ribose to D-ribose-5-phosphate, the entry metabolite of the non-oxidative pentose-phosphate pathway (PMID: 23651393). The adjacent ribonucleoside hydrolase (PP_2460) can additionally liberate ribose from nucleosides, feeding the same pathway.

Regulation. The LacI-family repressor RbsR (PP_2457) controls operon expression in a substrate-responsive manner. In *E. coli*, RbsR additionally links ribose availability to purine nucleotide synthesis regulation, a role that underscores the tight integration of ribose transport with central nucleotide metabolism (PMID: 23651393).

Division of labor / annotation clarity. Substrate specificity resides in RbsB, not RbsA-I. RbsA-I is the interchangeable energizing subunit; its function is transport energization, and its "substrate" in the enzymatic sense is ATP. The presence of a second K10441 paralog (PP_2759) that is *not* embedded in a complete catabolic operon reinforces that PP_2455 specifically is the ribose-catabolism-coupled importer ATPase.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
25533465	*In vitro reassembly of the ribose ABC transporter...*	**Primary experimental anchor.** Defines RbsA as the cytoplasmic ABC protein with fused NBDs (supports F001, F004); shows nucleotide-state-dependent assembly of $RbsABC_2$ / RbsAC complexes, evidencing ATP-hydrolysis-driven transport at the membrane (supports F003).
23651393	*Involvement of the ribose operon repressor RbsR in regulation of purine nucleotide synthesis in E. coli.*	Establishes the *rbsDACBK* operon organization and that RbsABC is the high-affinity D-ribose transporter feeding RbsD/RbsK to make D-ribose-5-phosphate (supports F002); links the operon to nucleotide metabolism via RbsR.
10941799	*Ribose utilization in Lactobacillus sakei...*	**Contrasting case.** Demonstrates that some bacteria dispense with the RbsABC ABC transporter entirely, using an alternative permease (RbsU) — highlighting that the ABC-transporter route seen in *P. putida* is one of several evolutionary solutions, and underscoring the value of *P. putida*'s complete *rbsBACR-KD* operon in confirming the ABC route here.

Assessment. The two supporting papers (PMID: 25533465 and PMID: 23651393) are precise, mechanism-focused studies of the *E. coli* ribose system — the closest well-characterized orthologue. Together with the 41% sequence identity and complete conserved operon in *P. putida*, they provide a robust, non-high-throughput basis for the functional assignment. No direct experimental characterization of PP_2455 itself (e.g., in vitro ATPase assay, transport assay, or knockout) was located; the assignment is therefore inference-by-orthology of high confidence rather than direct demonstration.

Limitations and Knowledge Gaps

1. **No direct experimental data on PP_2455 itself.** All mechanistic evidence derives from the *E. coli* orthologue. There is no published in vitro ATPase measurement, ribose-transport assay, structure, or gene-deletion phenotype specifically for *P. putida* RbsA-I. The assignment, while strong, is homology- and context-based.
2. **Substrate specificity is inferred, not measured, in *P. putida*.** D-ribose specificity is attributed to RbsB by analogy. It remains possible (though unlikely) that the *P. putida* system has broadened or shifted specificity toward a ribose-related pentose.
3. **Role of the second paralog (PP_2759 / rbsA-II) is unresolved.** Its cluster lacks co-localized ribokinase/pyranase; whether it transports ribose under different conditions, a different sugar, or feeds a different pathway is not established. This affects the completeness of the ribose-transport annotation for the organism.
4. **Regulatory details in *P. putida* are unverified.** RbsR-mediated, ribose-responsive control of the PP_2454–PP_2460 operon is inferred from *E. coli*; the effector and operator architecture in *P. putida* have not been experimentally mapped.
5. **Quantitative kinetics unknown.** Affinity (K_m/K_d) for ribose, ATP turnover rate, and coupling stoichiometry (ATP hydrolyzed per ribose imported) are unmeasured for this protein.

Proposed Follow-up Experiments / Actions

1. **Direct ATPase assay.** Heterologously express and purify RbsA-I (PP_2455), and measure Mg-ATP hydrolysis (malachite-green / NADH-coupled assay), ideally with and without reconstituted RbsC permease to demonstrate permease-stimulated ATPase activity.
2. **Transport assay in a defined background.** Construct a ΔPP_2455 deletion (and a ΔPP_2455 ΔPP_2759 double mutant) and assay ^{14}C-D-ribose uptake and growth on D-ribose as the sole carbon source. A growth/uptake defect rescued by complementation would provide direct in vivo proof of function and clarify the division of labor with the PP_2759 paralog.
3. **Reconstitution / structure.** Co-express RbsA-I with RbsC (PP_2456) and RbsB (PP_2454), reconstitute into nanodiscs/proteoliposomes, and use cryo-EM to capture nucleotide-state-dependent conformations, paralleling the *E. coli* study (PMID: 25533465).

4. **Specificity screen of RbsB.** Determine the binding spectrum of the periplasmic RbsB (PP_2454) by isothermal titration calorimetry against D-ribose and related pentoses to confirm that specificity is set at the binding-protein step.
5. **Operon regulation mapping.** Test RbsR (PP_2457) binding to the operon promoter by EMSA and identify the physiological inducer, confirming ribose-responsive transcriptional control in *P. putida*.
6. **Paralog functional dissection.** Characterize the PP_2757–PP_2761 cluster (including PP_2759) transport specificity to resolve whether it is a second ribose route or handles a distinct substrate, completing the transport annotation of the organism.

Conclusion

rbsA-I (PP_2455 / Q88K37) is the cytoplasmic ATP-binding subunit of the high-affinity D-ribose ABC importer of *Pseudomonas putida* KT2440. It is a 524-amino-acid, two-fused-NBD ABC-ATPase (EC 7.5.2.7, KEGG K10441) that hydrolyzes ATP at the cytoplasmic face of the inner membrane to energize D-ribose translocation through the RbsC permease, working with the periplasmic RbsB substrate-binding protein that dictates D-ribose specificity. It is embedded in a complete *rbs* operon (rbsB–rbsA-I–rbsC–rbsR–rbsK–rbsD–nucleoside hydrolase) that channels imported ribose, via ribokinase-produced D-ribose-5-phosphate, into the pentose-phosphate pathway. The assignment is supported by 41% identity to the biochemically characterized *E. coli* RbsA, full conservation of the dual Walker A/Walker B catalytic apparatus, unambiguous operon context, and reconstitution studies of the *E. coli* $RbsABC_2$ transporter — while remaining, at present, an inference-by-orthology that has not yet been directly validated for the *P. putida* protein itself.
